Throughput Engineering

CURRENT BASELINE & PROPOSED SOLUTIONS

1. Current Throughput Baseline

Based on the 7-day realistic operational dataset, the fleet's current performance metrics establish our optimization baseline.

5,000	132.8	37.64
TOTAL TASKS COMPLETED	TOTAL ACTIVE HOURS	CURRENT TASKS / HOUR

2. Proposed Solutions

Derived from the hypothesis validation phase, three targeted orchestration updates have been identified to reclaim lost capacity.

Solution 1: Predictive & Opportunistic Charging (Fixes H4)

Replaces the 24% reactive charging threshold. Forces robots to charge during identified `Idle` windows, preventing battery depletion during peak WMS queuing hours.

Solution 2: Proximity-Weighted Routing (Fixes H1)

Deprecates strict FIFO logic. Evaluates robot spatial coordinates (X/Y) relative to pickup zones to drastically reduce the current 25.7% empty travel cycle loss.

Solution 3: Zone-Based Traffic Corridors (Fixes H3)

Implements one-way digital routing through Dispatch and Aisle 1 to alleviate the 61% spatial congestion bottleneck found in high-density staging areas.

3. Impact Projection: Solution 1 (Predictive Charging)

Calculates the throughput yield of shifting peak-hour charging into standard idle windows.

Step 1: Identify peak operational charging loss (8:00 AM – 6:00 PM).
Result: `133.75 hours`

Step 2: Assume a conservative 50% efficiency rate in forcing this charging into the existing 139.8 hours of global idle time.
Result: 133.75 × 0.50 = `66.88 active hours reclaimed`

Step 3: Convert reclaimed hours into net new tasks using the baseline cycle time (465s / 0.1291h).
Result: 66.88h / 0.1291h = `+518 new tasks`

Step 4: Calculate updated throughput: (5,000 baseline + 518) / 132.83 hours.
New Throughput: `41.54 tasks / hour` (+10.4% increase)

4. Impact Projection: Solution 2 (Proximity Routing)

Calculates the throughput yield of optimizing task dispatch to reduce empty travel time.

Step 1: Identify current empty travel loss. Average empty travel is 119.4s out of a 465s total cycle.

Step 2: Apply optimization logic. Assume proximity routing reduces empty travel by 50% (saving 59.7s per task).
Result: New Cycle Time = 465s - 59.7s = `405.3s`

Step 3: Calculate active capacity. The fleet currently spends 645.8 robot-hours actively executing tasks (5,000 tasks × 465s). How many 405.3s tasks fit into this active budget?
Result: 645.8h / (405.3s / 3600) = `5,736 total tasks`

Step 4: Calculate updated throughput: 5,736 tasks / 132.83 total hours.
New Throughput: `43.18 tasks / hour` (+14.7% increase)

5. Validation of the 50% Conversion Assumption

The projection for Solution 1 relies on the mathematical assumption that **50% of peak-hour charging** can be successfully shifted into existing idle windows. This is a heuristic estimate; physical and operational realities will dictate the true conversion rate.

Recommended Validation Strategies

To derisk this assumption before a fleet-wide deployment, I recommend the following validation phases:

- **Digital Twin Simulation (Shadow Mode):**
  Run the predictive charging algorithm against historical `Fleet_State_Log` data. Calculate the exact delta in battery levels if robots had routed to chargers during their recorded idle states.
- **Hardware Constraint Auditing:**
  Verify the physical charge-acceptance rate of the robots. Ensure that a short 3-minute idle window provides a mathematically meaningful battery increase (e.g., +2%) rather than being entirely lost to docking and undocking overhead.
- **Partial Fleet Pilot (Canary Release):**
  Deploy the updated charging threshold solely to the conveyor class robots (CV-01 through CV-05) for a 48-hour period. Measure the localized throughput delta against the forklift class before pushing the update facility-wide.